1. Friction

1.1. Introduction

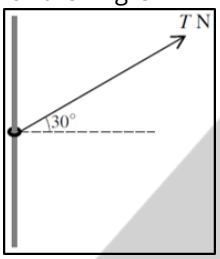
$$\text{Friction} = \text{Coefficient of Friction} \times \text{Contact Force}$$

$$F = \mu r$$

- Friction always acts in the opposite direction of motion
- Limiting equilibrium: on the point of moving, friction at max (limiting friction)
- Limiting equilibrium is also used to describe a situation when the object is about to slip, where friction is at its max
- Smooth contact: friction negligible
- Contact Force:
  - In situations where the particle is on a horizontal plane, the contact force or reaction force is equal to the weight of the particle (mg)
  - in situations where the particle is on an inclined plane, the contact force or reaction force is equal to the verticle component of the weight of the particle ($mgcos(\theta)$)

{W11-P43} Question 6:

The ring has a mass of $2kg$. The horizontal rod is rough, and the coefficient of friction between the ring and rod is 0.24. Find the two values of T for which the ring is in limiting equilibrium



Solution

The ring is in limiting equilibrium in two different scenarios; we have to find T in both:

Scenario 1: the ring is about to move upwards

$$Resultant = T\sin 30 - friction - Weight\ of\ Ring$$

Since the system is in equilibrium, resultant = 0:

$$Contact\ Force = T\cos 30$$

$$\therefore Friction = 0.24 \times T\cos 30$$

Substitute relevant information into to initial equation

$$0 = T\sin 30 - 0.24T\cos 30 - 20$$

$$T = 68.5N$$

Scenario 2: the ring is about to move downwards

This time, friction acts in the opposite direction since friction opposes the direction of motion, thus:

$$Resultant = T\sin 30 + Friction - Weight\ of\ Ring$$

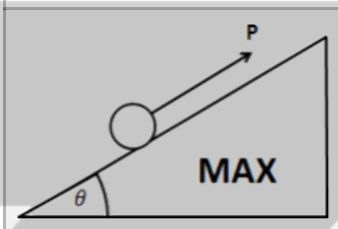
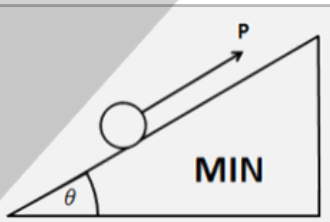
Using information from before:

$$0 = T\sin 30 + 0.24T\cos 30 - 20$$

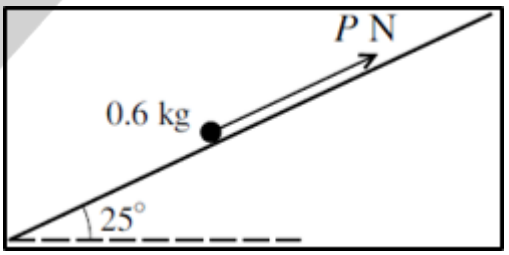
$$T = 28.3N$$

1.2. Equilibrium

Force required to keep a particle in equilibrium on a rough plane

Max Value	Min Value
	
The particle is about to move upThus, friction force acts down the slope $P = F + mgsin\theta$	The particle is about slip downThus, frictional force acts up the slope$F + P = mgsin\theta$

{W12-P43} Question 6:



Coefficient of friction is 0.36 and the particle is in equilibrium. Find the possible values of P

Solution

The magnitude of friction on particle in both scenarios is the same but acting in opposite directions

Calculate the magnitude of friction first:

$$Contact\ Force = 6\cos 25$$

$$\therefore Friction = 0.36 \times 6\cos 25$$

Scenario 1: particle is about to move upwards

$$P = 6\sin 25 + Friction$$

$$P = 4.49N$$

Scenario 2: particle is about to move downwards

$$P = 6\sin 25 - friction$$

$$P = 0.578N$$

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